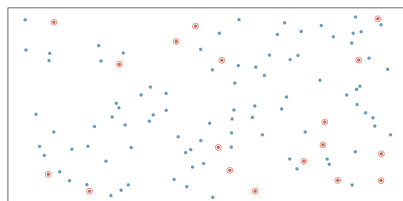
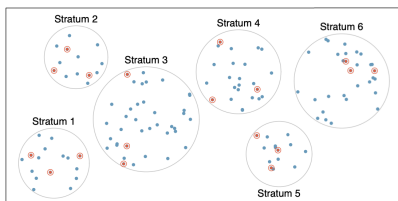


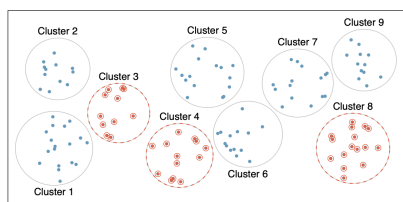
# STATISTICS & PROBABILITY GUIDE



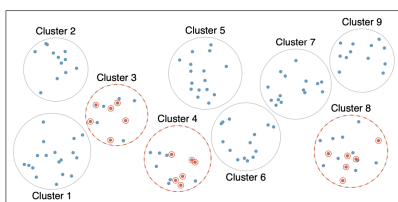
A simple random sample.



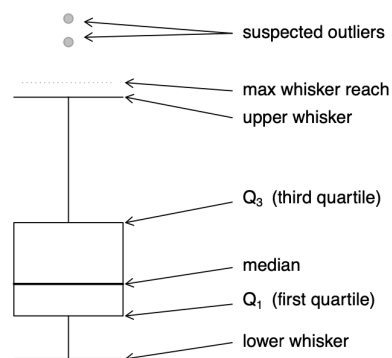
A stratified sample.



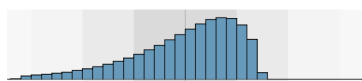
A clustering sample.



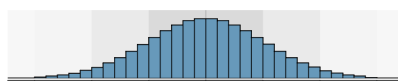
A multistage clustering sample.



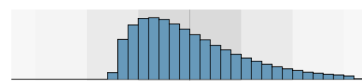
The features of a boxplot.



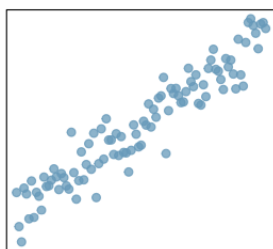
A left-skewed distribution.



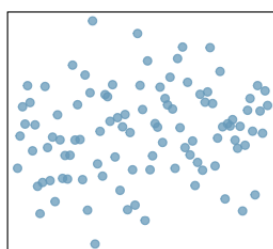
A symmetric distribution.



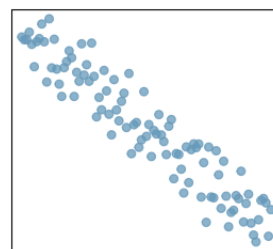
A right-skewed distribution.



A positive linear association.



No association.



A negative linear association.

Association is *not* causation.

*Observational studies* do not interfere with the data.

*Experimental studies* use *treatment* and *control* groups to reduce influence of *confounding variables*.

*Generalization* requires a *representative sample* to reduce *bias*.

*Causal inference* requires an *experimental study*.

# STATISTICS & PROBABILITY GUIDE

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Median is the 50th percentile.

$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}}$$

$$IQR = Q_3 - Q_1$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$P(A) = 1 - P(A^c)$$

$$P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$$

$$P(A \text{ and } B) = P(A) \cdot P(B | A)$$

$$P(k \text{ successes out of } n \text{ trials}) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k! \cdot (n - k)!}$$

$$\mu = E(X) = x_1 \cdot P(X = x_1) + x_2 \cdot P(X = x_2) + \cdots + x_n \cdot P(X = x_n)$$

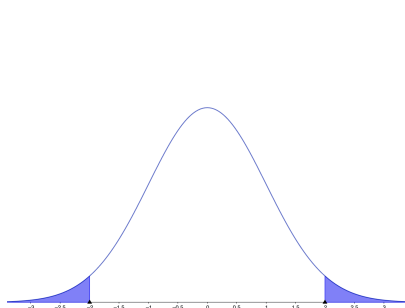
$$\text{Var}(X) = (x_1 - E(X))^2 \cdot P(X = x_1) + (x_2 - E(X))^2 \cdot P(X = x_2) + \cdots + (x_n - E(X))^2 \cdot P(X = x_n)$$

$$\sigma = \sqrt{\text{Var}(X)}$$

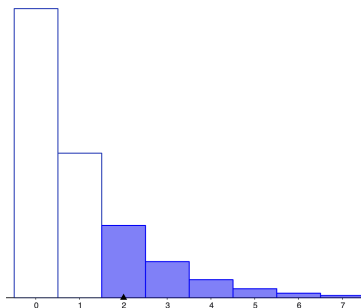
$$E(aX + bY) = a \cdot E(X) + b \cdot E(Y)$$

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + b^2 \cdot \text{Var}(Y)$$

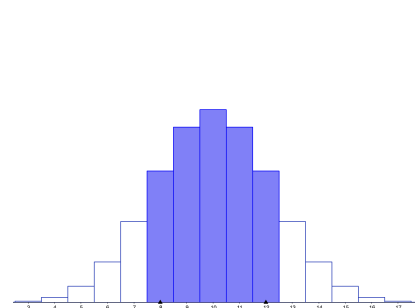
Probability can be thought of in terms of *relative frequency* as well as *area under a distribution*.



$P(|Z| \geq 2)$  shaded on a standard normal distribution,  $\mu = 0$ ,  $\sigma = 1$ .



$P(X \geq 2)$  shaded on a geometric distribution,  $\mu = 1/p$ ,  $\sigma = \sqrt{1 - p}/p$ .



$P(6 \leq X \leq 12)$  shaded on a binomial distribution,  $\mu = np$ ,  $\sigma = \sqrt{np(1 - p)}$ .